Half adder and Full Adder

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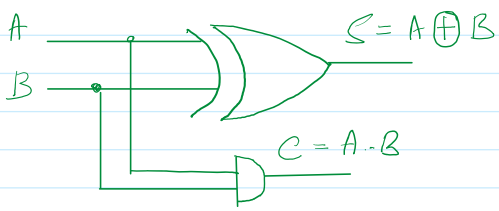
Half adder and a full adder are fundamental building blocks of digital electronic circuit, used to perform binary addition. They form the basis of arithmetic operations inside the computer and digital system.

Half adder

Half adder is a combinational logic circuit that is designed by connecting one XOR gate and one AND gate. The half adder circuit has two inputs: A and B which adds two input digits and generate a carry and sum as output.

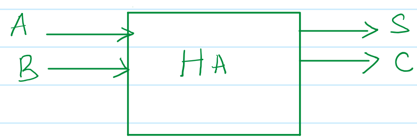


logic ckt.

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

XOR AND

Truth table



Block diagram

H.W 1. Draw HA using basic gates only (AND, OR, NOT)

2. Draw HA using Universal gates only (i) using NOR gates only

(ii) Using NAND gates only

Applications

- Arithmetic operations like addition, subtraction and multiplication in low level dynamic circuits.
- Three types of rectifiers: half wave, full wave and full wave with center tabbed secondary. Used in a small integrated circuits.

Advantages:

- Flexible and easy when it comes to design.
- Involves the use of fewer logic gates, thus is cheaper.

Disadvantages

- Fails to process a carry input from the previously added numbers.
- Restricted to the addition of only two bits.

Full adder

Full adder is the combinational circuit that consist of 1-XOR gates and three AND gates and one OR gate. Full adder is the adder that adds 3

Full adder

Full adder is the combinational circuit that consist of 1-XOR gates and three AND gates and one OR gate. Full adder is the adder that adds 3 inputs and outputs Sum and Carry.

$[A + A = A]$

$[A + \bar{A} = 1]$

$$C = \bar{X}YZ + X\bar{Y}Z + XY\bar{Z} + XYZ$$

$$= \bar{X}YZ + XYZ + X\bar{Y}Z + XYZ + XY\bar{Z} + XYZ$$

$$= YZ(\bar{X} + X) + XZ(\bar{Y} + Y) + XY(\bar{Z} + Z)$$

$$C = YZ + XZ + XY$$

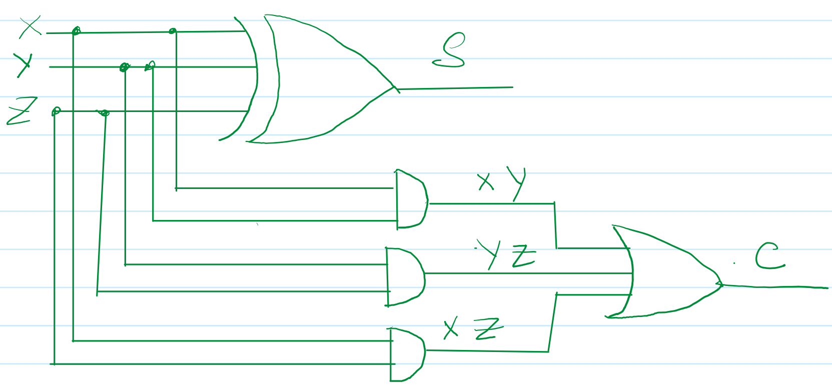
X	Y	Z	S	C	
0	0	0	0	0	$X + Y + Z$
0	0	1	1	0	$X + Y + \bar{Z}$
0	1	0	1	0	$X + \bar{Y} + Z$
0	1	1	0	1	$\leftarrow \bar{X}YZ$
1	0	0	1	0	$\bar{X} + Y + Z$
1	0	1	0	1	$\leftarrow X\bar{Y}Z$
1	1	0	0	1	$\leftarrow XY\bar{Z}$
1	1	1	1	1	$\leftarrow XYZ$

Truth table XOR

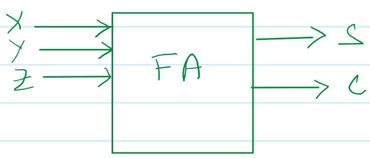
$$\boxed{\begin{aligned} S &= (X \oplus Y) \oplus Z \\ C &= XY + YZ + XZ \end{aligned}}$$

$\leftarrow$ SOP

dual $= (X + Y) \cdot (Y + Z)(X + Z)$

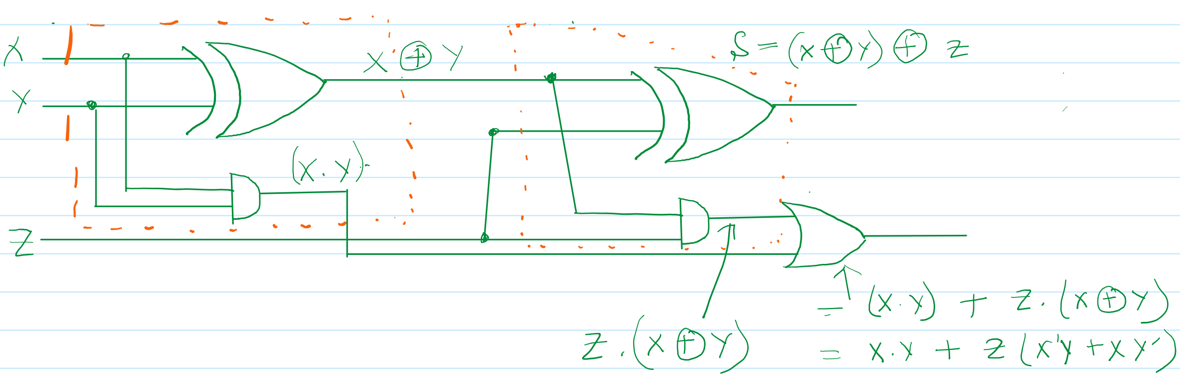


Logic ckt.



Block diagram.

Full adder using two half adders:



$Z\cdot(X \oplus Y)$

$$
\begin{aligned}
&= (X\cdot Y) + Z\cdot(X \oplus Y) \\
&= X\cdot Y + Z(X'Y + XY') \\
&= X\cdot Y + X'YZ + XY'Z \\
&= Y(X + \bar{X}Z) + XY'Z \\
&= Y(X+\bar{X})(X+Z) + XY'Z \\
&= XY + YZ + XY'Z \\
&= XY + Z(Y + XY') \\
&= XY + Z(Y+X)(Y+Y') \\
C &= XY + YZ + XZ
\end{aligned}
$$

Homework:

Draw Full adder 1. Using ~~basic~~ gates only (AND, OR, NOT)

2. Using NOR gates only.

3. Using NAND gates only.

[Try to use minimum number of gates]

Advantages

- Can add 3 bits, it includes one carry input and a carry output which can perform more elaborate computations.
- It can be cascaded to produce adders for a number of bit additions which make it suitable for multi-bit arithmetic.

Disadvantages

- Complex and needs more gates, hence making the design more complicate and expensive.
- Slightly slower because normally two gate process used instead of one.

Applications

- Present in the arithmetic logic units (ALU) and other complicated digital systems.

- Carry look ahead adders and digital processors that utilize multi-bit binary addition.